

Introduction to Electrical Engineering

What electrical engineering is

Electrical engineering deals with electricity in both theory and practice: generating it, transmitting it, controlling it, measuring it, and using it in devices and machines. It connects physics, mathematics, electronics, energy systems, automation, and computation, so it is one of the most fundamental engineering branches.

The subject is not only about wires and circuits. It also includes power systems, control systems, electronics, telecommunications, signal processing, instrumentation, and computer hardware.

Core purpose

The central purpose of electrical engineering is to move and control energy and information using electrical signals. That is why EE is involved in everything from household wiring and motor drives to mobile networks, renewable energy, embedded systems, and industrial automation.

Main divisions

Electrical engineering is usually divided into several major areas:

- Power engineering.
- Control engineering.
- Electronics engineering.
- Telecommunications engineering.
- Signal processing.
- Instrumentation engineering.
- Computer engineering.

Power engineering focuses on generation, transmission, distribution, and utilization of electric power. Control engineering focuses on Modeling and controlling dynamic systems. Electronics engineering deals with circuits and devices, while telecommunications handle information transfer over communication channels.

Basic electrical quantities

The first unit of electrical engineering usually introduces the quantities every student must know:

- Electric charge.
- Electric current.
- Voltage.
- Resistance.
- Power.
- Energy.

Current is the flow of electric charge, measured in amperes. Voltage is electric potential difference, measured in volts. Resistance opposes current flow, and power is the rate at which electrical energy is transferred or consumed.

Fundamental laws

Three laws form the backbone of basic circuit analysis:

- Ohm's law.
- Kirchhoff's Current Law.
- Kirchhoff's Voltage Law.

Ohm's law relates voltage, current, and resistance through $V = IR$. Kirchhoff's Current Law expresses conservation of charge at a node, and Kirchhoff's Voltage Law expresses conservation of energy around a loop. These laws are essential for Analyzing any electrical network.

Circuit elements

In introductory EE, students learn ideal and practical circuit components:

- Resistors.
- Capacitors.
- Inductors.
- Sources of voltage and current.
- Switches.
- Loads.

These elements combine to form DC and AC circuits, filters, timing networks, and many real engineering systems.

AC and DC basics

Direct current flows in one direction, while alternating current changes direction periodically. AC is dominant in power systems because it is easy to transform to different voltage levels, while DC is common in electronics, batteries, and digital systems.

Students also study RMS value, frequency, phase difference, impedance, and power factor when they move from simple DC circuits to AC analysis.

Electrical measurements

Measurement is a major part of EE because engineers must verify voltage, current, resistance, power, frequency, and waveform behavior. Common instruments include voltmeters, ammeters, multimeters, oscilloscopes, wattmeter, and clamp meters.

Good measurement practice matters because incorrect testing can damage equipment, create safety hazards, or lead to wrong design conclusions.

Machines and power systems

A major part of EE is the study of electrical machines and power systems. Transformers, generators, motors, transmission lines, and distribution systems form the infrastructure that delivers electricity from power plants to end users.

This is why electrical engineering is critical for energy systems, renewable integration, electric mobility, and industrial production.

Electronics and semiconductors

Electrical engineering overlaps strongly with electronics, especially in diodes, transistors, rectifiers, amplifiers, sensors, and integrated circuits. Semiconductor devices make signal processing, digital logic, embedded systems, and modern communication possible.

At an introductory level, students should understand the difference between power-focused electrical systems and signal-focused electronic systems.

Control and automation

Control engineering studies how to make systems behave in a desired way using feedback. Examples include motor speed control, robot motion, temperature regulation, and aircraft autopilot systems.

This area is important because modern electrical systems increasingly combine sensors, controllers, actuators, and software.

Communication and signals

Telecommunications and signal processing deal with how information is represented, transmitted, filtered, compressed, and reconstructed. Electrical engineers work with analog and digital signals in audio, video, wireless communication, imaging, and data networks.

BrainStack

This is one reason EE is central to the internet, smartphones, radar, satellites, and modern wireless systems.

Why every engineer studies it

Even if a student is not majoring in electrical engineering, the subject is common to all branches because:

- Every machine needs power.
- Every industrial plant uses electrical control.
- Every modern system uses sensors and electronics.
- Every engineer must understand safety and energy use.

It also gives students a systems mindset: how energy enters a device, how it is controlled, and how it becomes useful work.

1. Scope of electrical engineering

Electrical engineering studies electric charge, current, voltage, power, circuits, machines, control, communication, and electromagnetism. It covers power generation and transmission, electronics, automation, instrumentation, telecommunications, and computer-related hardware.

Full syllabus map

- Basics of electrical quantities and units.
- Circuit elements and network laws.
- DC circuits.
- AC fundamentals.
- Magnetic circuits and electromagnetism.
- Electrical machines.
- Power generation, transmission, and distribution.
- Measuring instruments.
- Electronics and semiconductor devices.
- Control systems.
- Communication and signal basics.

2. Basic electrical quantities

These are the building blocks of the subject:

- Charge Q , measured in coulombs.
- Current I , measured in amperes.
- Voltage V , measured in volts.
- Resistance R , measured in ohms.
- Power P , measured in watts.
- Energy W or E , measured in joules or watt-hours.

Key formulas

- $I = \frac{Q}{t}$
- $V = \frac{W}{Q}$
- $P = VI$

- $E = Pt$

3. Ohm's law and circuit basics

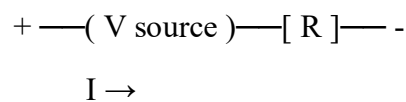
Ohm's law relates voltage, current, and resistance in a conductor under constant physical conditions. It is one of the most important laws in introductory electrical engineering.

Key formulas

- $V = IR$
- $R = \rho \frac{L}{A}$
- $P = I^2 R$
- $P = \frac{V^2}{R}$

Simple circuit diagram

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4. Kirchhoff's laws

Kirchhoff's laws are used to analyze complex circuits.

Kirchhoff's Current Law

At any node, total current entering equals total current leaving.

$$\sum I = 0$$

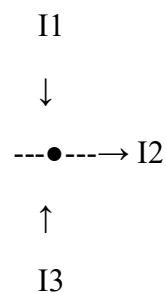
Kirchhoff's Voltage Law

The algebraic sum of voltages around any closed loop is zero.

$$\sum V = 0$$

Node and loop idea

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5. DC circuits

DC circuits deal with constant voltage and current. Common topics include series and parallel resistive circuits, source transformation, mesh analysis, node analysis, and power calculations.

Series circuits

- Same current flows through all components.
- Total resistance:

$$R_T = R_1 + R_2 + R_3 + \dots$$

Parallel circuits

- Same voltage across branches.
- Total resistance:

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

Voltage divider

$$V_{out} = V_{in} \frac{R_2}{R_1 + R_2}$$

6. AC fundamentals

Alternating current changes direction periodically and is the standard for power systems. Students learn waveform shape, frequency, phase, RMS values, reactance, impedance, and power factor.

Key formulas

- $v(t) = V_m \sin(\omega t)$
- $i(t) = I_m \sin(\omega t + \phi)$
- $V_{rms} = \frac{V_m}{\sqrt{2}}$
- $I_{rms} = \frac{I_m}{\sqrt{2}}$
- $\omega = 2\pi f$

AC wave sketch

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V

| / \ / \
| / \ / \
|----/-----\--/-----\----> t
| / \ \

Impedance

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Reactance

- Inductive reactance:

$$X_L = \omega L$$

- Capacitive reactance:

$$X_C = \frac{1}{\omega C}$$

Power in AC

- $P = VI \cos \phi$
- Power factor = $\cos \phi = \frac{R}{Z}$

7. Magnetism and electromagnetic induction

Magnetism is the foundation of motors, generators, transformers, and many industrial devices. This area connects electricity and magnetism through moving charges and changing magnetic fields.

Key formulas

- Magnetic force on charge:

$$F = qvB \sin \theta$$

- Magnetic force on conductor:

$$F = BIL \sin \theta$$

- Magnetic flux:

$$\Phi = BA \cos \theta$$

- Faraday's law:

$$e = -\frac{d\Phi}{dt}$$

- Self-induction:

$$e = -L \frac{di}{dt}$$

Simple induction idea

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Changing flux $\rightarrow$ Induced emf $\rightarrow$ Current

8. Transformers

Transformers transfer AC power between circuits by electromagnetic induction and are essential in power systems.

Key formulas

- Turns ratio:

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$

- Current ratio:

$$\frac{I_1}{I_2} = \frac{N_2}{N_1}$$

- Ideal power relation:

$$V_1 I_1 = V_2 I_2$$

Simple transformer diagram

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AC in $\rightarrow$ [Primary coil || Iron core || Secondary coil] $\rightarrow$ AC out

9. Electrical machines

Electrical machines convert electrical energy to mechanical energy or the reverse. The main types are DC machines, transformers, induction motors, synchronous machines, and special-purpose motors.

Main ideas

- Generator: mechanical to electrical.
- Motor: electrical to mechanical.
- Transformer: AC voltage level change without mechanical motion.

Motor principle

A current-carrying conductor in a magnetic field experiences force. This is the basic operating principle of electric motors.

10. Power system basics

Power engineering covers generation, transmission, distribution, protection, and utilization of electrical power. It is one of the largest branches within electrical engineering.

Core topics

- Generation sources: thermal, hydro, nuclear, solar, wind.
- Transmission lines.
- Distribution networks.
- Substations.
- Switchgear and protection.
- Earthing and safety.

Typical power flow

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Power plant → Step-up transformer → Transmission lines → Substation → Distribution → Consumer

11. Electrical measurements

Measurement is critical because engineers must verify current, voltage, power, resistance, frequency, and waveform quality. Instruments include ammeter, voltmeter, multimeter, wattmeter, energy meter, oscilloscope, and power factor meter.

Important principles

- Ammeter is connected in series.
- Voltmeter is connected in parallel.
- Ideal ammeter has zero resistance.
- Ideal voltmeter has infinite resistance.

12. Electronics and semiconductor devices

Electrical engineering includes basic electronics such as diodes, transistors, rectifiers, amplifiers, and logic devices. Semiconductor devices are the backbone of modern circuits and digital systems.

Key topics

- PN junction diode.
- Rectifier.
- Zener diode.
- BJT.
- FET.
- Basic logic gates.

Basic diode idea

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P-type | junction | N-type

→ current easier in forward bias

13. Control systems

Control engineering studies how to make systems behave in a desired manner using feedback. It appears in robotics, industrial automation, process control, vehicles, aircraft, and smart devices.

Main ideas

- Open-loop and closed-loop systems.
- Feedback.
- Stability.
- Transfer function.
- Step response.
- PID control basics.

14. Communication and signal basics

Electrical engineers also deal with signals and communication. Topics include analog vs digital signals, modulation, bandwidth, noise, and transmission basics.

Core concepts

- Signal.
- System.
- Sampling.
- Modulation.
- Noise.
- Channel capacity basics.

15. Important derivations and formula sheet

DC

- $V = IR$
- $R_T = R_1 + R_2 + \dots$
- $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

Power

- $P = VI$
- $P = I^2 R$
- $P = \frac{V^2}{R}$

AC

- $V_{rms} = \frac{V_m}{\sqrt{2}}$
- $I_{rms} = \frac{I_m}{\sqrt{2}}$
- $Z = \sqrt{R^2 + (X_L - X_C)^2}$
- $X_L = \omega L$
- $X_C = \frac{1}{\omega C}$

Magnetism and induction

- $F = qvB \sin \theta$
- $F = BIL \sin \theta$
- $\mathcal{E} = -\frac{d\Phi}{dt}$
- $\mathcal{E} = -L \frac{di}{dt}$

Transformer

- $\frac{V_1}{V_2} = \frac{N_1}{N_2}$
- $\frac{I_1}{I_2} = \frac{N_2}{N_1}$

16. Exam-style diagram list

You should practice drawing these:

- Basic electrical circuit.
- Series circuit.
- Parallel circuit.
- AC sine wave.
- Transformer.
- Magnetic field around conductor.
- Node and mesh diagrams.
- Basic diode symbol and biasing.
- Motor principal sketch.
- Power system block diagram.

17. How to study this subject

First learn quantities, units, and basic laws. Then move into circuit analysis, AC concepts, magnetism, machines, measurement, and finally electronics and control. This sequence helps because each topic builds on the previous one.